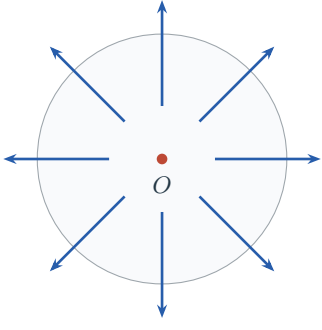
M1-22 电势的径向斜率与电场方向相反

图示 $q > 0$，电场向外



$q < 0$ 时电场箭头全部反向

$$\varphi = \frac{q}{4\pi\varepsilon_0 r},\ r > 0$$

$$\nabla\varphi = -\frac{q}{4\pi\varepsilon_0 r^2}\hat{\mathbf{r}}$$

$$\mathbf{E} = -\nabla\varphi = \frac{q}{4\pi\varepsilon_0 r^2}\hat{\mathbf{r}}$$